

Apply for a Science Corps Fellowship - Entry #3247

Name

Rachele Ossola

Emailrachele.ossola@colostate.edu**Gender**

Female

City, state, and country of residence

Fort Collins, Colorado, United States

Nationality

Italy

When did or will you finish your PhD?

2021

University Education

Ph.D. | Environmental Chemistry | ETH Zurich, Switzerland | 2021

Master | Biological and Molecular Chemistry | EPFL, Switzerland | 2016

Bachelor | Chemistry | University of Milan, Italy | 2014

What are your reasons for applying to be a Science Corps fellow?

After the end of my postdoctoral fellowship, I will take a "sabbatical" to expand my expertise in citizen science, teaching, and outreach and to get in contact with realities that may inspire research themes and initiate future partnerships. The Science Corp fellowship is the perfect opportunity to meet these goals. In particular, I am excited to collaborate with teachers to develop educational material and targeted citizen science projects. As an environmental chemist, I am also keen on learning about the specific environmental problems of these communities. Lack of expertise and/or scientific

equipment is sometimes the underlying cause of unresolved environmental issues in developing countries. Citizen science projects may contribute to building know-how and the foundations for future research partnerships with local schools and universities.

In addition to your studies, describe any relevant experiences that have prepared you for a Science Corps fellowship.

My passion for science communication, outreach, teaching, and my international background make me well suited to become a Science Corp fellow. The three years at a popular science center during high school is perhaps the most relevant experience. Among many other activities, I led weekly guided visits to the astronomical observatory and the botanical garden, co-organized and actively participated in outreach events for schools and the general public, and co-developed educational material (including a brand-new website for the botanical garden). This immersive experience taught me how to be an engaging speaker and the power of sharing my passion for science. After moving to a German-speaking country, where outreach was "out-of-reach", I shifted my focus to university teaching. I co-designed and led a laboratory course on environmental photochemistry (that also resulted in a scientific publication) and helped re-design a first-year chemistry lab course, in addition to the regular graduate students' teaching duties. During the last few years, I also occasionally translated educational material for the International Younger Chemistry Network, created outreach documents for the United Nations Environmental Program, and went to schools to talk about my research.

Where do you imagine yourself professionally in the future? Check all that apply.

Academic scientist in academia

In science education

In science outreach

Expand on your selection above and describe where you imagine yourself professionally in the near (two year) and long-term (ten year) future.

My long-term goal is to return to Italy, my home country, and establish a cutting-edge research program in environmental chemistry that connects the "purely academic world" to the layperson. On the academic side, I will continue exploring the oxidation chemistry of the complex mixture of chemicals ubiquitous in the environment – aerosols in the air and "organic matter" in soil, water, and natural surfaces. In addition to this cutting-edge topic, I will initiate a new research theme focused on low-cost chemistry instruments for educational purposes. While we will build and test these instruments in an academic lab to ensure data quality, citizen scientists will employ them in environmental science projects led by my research team. These projects will be the starting point for creating a popular science center like the one I frequented as a teenager. Although I am committed to Italy, these initiatives are suitable for replication in developing countries. During my sabbatical (2024/25), I will put my research on hold to gain experience in teaching and outreach in underserved communities – through Science Corps and other organizations. This experience will give me the opportunity to build a network in these countries and expand my research and outreach efforts beyond Italy.

CV/Resume



[RacheleOssola CV.pdf](#)

How did you find out about Science Corps?

University email list

Any other comments/questions?

Favorite site: Morocco; starting date: April 2024.

(I am also open to going to other sites, with the same starting date)

[Science Corps](#)